

## HELIOLITE · DESIGN DECISION RECORD

# HelioLite TRL 2 review decisions

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## DOCUMENT CONTROL

|             |                                                        |         |                |
|-------------|--------------------------------------------------------|---------|----------------|
| Document ID | HLT-DDR-001                                            | Version | 0.2            |
| Status      | <b>Draft</b>                                           | Date    | 2026-09-25     |
| Author      | Amish Chadha                                           | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept                          |         |                |
| Repository  | github.com/BoujeeEnjinia1701/heliolite                 |         |                |
| Source      | docs/decisions/0001-trl2-review-decisions.md @ 091f03f |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                            |
|---------|------------|--------------|-----------------------------------------------------------------------------------|
| 0.2     | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002)                                       |
| 0.1     | 2026-09-25 | Amish Chadha | Record Amish's decisions on the TRL 2 review items and the items that remain open |

## 0001: TRL 2 review decisions

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- **Date:** 2026-09-25
- **Status:** accepted for items D1 to D10; item O4 decided in HLT-DDR-002; items O1 to O3 remain proposed

### ■ Context

The TRL 2 review note ( docs/REVIEW.md , session 2026-09-25, /populate) listed eleven items as "Proposed, awaiting Amish". Ten carried a recommendation. On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." Every item that carried a recommendation is therefore decided in favor of that recommendation. Items without a recommendation stay open.

The same instruction approved portfolio-wide decisions on the SwapCell interface (a wake method for hosts without CAN, a charge-while-discharging mode and a latch vibration rating) and on pricing shared SwapCell packs once. HelioLite runs from a listed indoor 12 V adapter and uses no SwapCell pack, so these do not change this design. The instruction also leaves co-design partners open for community designs; HelioLite's first site and user (O1) stays open accordingly.

### ■ Options considered

The options for each item are those listed in docs/REVIEW.md (TRL 2 session) and in HLT-PRC-001 v0.2, Key design choices.

### ■ Decision

*Table 1. Decided items.*

| #   | Item                           | Decision                                                                                                                                                                                                                   |
|-----|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D1  | Mast                           | 60.3 mm (2 in) galvanized steel pipe (Option A), rather than an 80 x 80 mm extrusion or a guyed 40 x 40 mm extrusion. Decided by Amish, 2026-09-25: go with recommendation.                                                |
| D2  | Storm awareness (R9)           | Wi-Fi wind forecast plus a low-cost cup anemometer (Options A and B), with the anemometer counted as a weather sensor, not a tracking sensor, and costed in the BOM. Decided by Amish, 2026-09-25: go with recommendation. |
| D3  | Safe state on power loss (R10) | Supercapacitor stow reserve that finishes a face-down stow (Option A). Decided by Amish, 2026-09-25: go with recommendation.                                                                                               |
| D4  | Budget                         | Raise <code>budget_usd</code> from \$400 to \$430 (option a), because the fixes for R9 and R10 are safety items. Decided by Amish, 2026-09-25: go with recommendation.                                                     |
| D5  | Pitch wording                  | Change "with no sensors" to "with no sun sensors". Decided by Amish, 2026-09-25: go with recommendation.                                                                                                                   |
| D6  | Stow attitude                  | Face-down stow at night, on faults and before storms, rather than face-up or edge-on. Decided by Amish, 2026-09-25: go with recommendation.                                                                                |
| D7  | Calibration                    | Jog the sun spot onto the target at three or more times of day and fit the mount model, rather than a compass, inclinometer or GPS survey. Decided by Amish, 2026-09-25: go with recommendation.                           |
| D8  | Sun-position algorithm         | Grena algorithm 5 at 30 s updates, NREL SPA as an alternative, with weekly network time when Wi-Fi is available. Decided by Amish, 2026-09-25: go with recommendation.                                                     |
| D9  | Power supply                   | Listed indoor 12 V adapter with only SELV cable outdoors, rather than a solar panel and battery on the mast. Decided by Amish, 2026-09-25: go with recommendation.                                                         |
| D10 | Mirror                         | Glass mirror with a safety backing film, rather than acrylic or aluminum-film mirror. Decided by Amish, 2026-09-25: go with recommendation.                                                                                |

#### Notes on the decided items:

- **D4.** `project.yaml` now carries `budget_usd: 430`, and requirement R15 reads "\$430 or less". The TRL 3 BOM totals \$425 (HLT-CAL-001, section J).
- **D5.** Applied to `project.yaml` and `README.md`. The design uses two axis homing Hall switches and, after D2, an anemometer; neither senses the sun.
- **D7.** HLT-CAL-001 (section D) finds that three points over about 3 h leave a 95th percentile residual of about 0.45° on the mirror normal, and four points over about 4 h about 0.23°. The decision allows "three or more"; four points over about 4 h are used in the pointing budget.

Table 2. Items that remained open after this record (O1 to O3 still Proposed, awaiting Amish).

| #  | Item                                                                                                                                                                         | Why it stays open                                                                            |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| O1 | First site and user for the co-design checklist (house, greenhouse or school)                                                                                                | No recommendation was made; needs Amish. Co-design partners stay open by Amish's instruction |
| O2 | Local rules on glare and structures at the first site                                                                                                                        | Depends on O1                                                                                |
| O3 | Siting survey of real yards (how often a mirror can see both the winter sun and a north window)                                                                              | Listed as an open question with no recommendation                                            |
| O4 | New TRL 3 proposals: stowed wind survival (R9), mass limit (R13), drive preload (R4), calibration time wording (R7), transient beam path in a stow (R10), calibration points | Decided by Amish, 2026-09-25: go with recommendation. Recorded in HLT-DDR-002 (N1 to N6)     |

## ■ Consequences

- The BOM gains an anemometer (line 14) and a supercapacitor stow reserve (line 15); the drives become NEMA17 motors on NMRV030-class 50:1 worm gearboxes with a listed rating, so the holding torque question in R9 can be checked.
- HLT-PRB-001, HLT-PRC-001 and HLT-REQ-001 move to v0.3 with these choices recorded as decided.
- TRL 4 work stays on hold by Amish's instruction.